



Modeling temperature gradients in wine fermentation tanks

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ABSTRACT

Extreme temperatures are common in large wine fermentation tanks. If not controlled properly, they can lead to problematic fermentations. Thus, efficient cooling and automatic control systems must be designed. However, it is rather difficult to design and implement effective cooling and control systems without adequate models able to reproduce the complex dynamic behavior observed in large fermentors. Therefore, we developed a compartmental dynamic mass-and-energy-balance model able to simulate temperature and concentration gradients in large wine fermentation tanks. This paper presents the model, along with all its parameters. It discusses simulations of temperature, alcohol content, density, biomass, and sugar concentrations. Finally, it presents a sensitivity analysis of the must temperature dynamics. The model reproduced reasonably well the values of the observed variables, including the most critical one: must temperature (with an absolute mean error of 1.4 °C). After proper calibration, it can be used to design control strategies for cooling in large wine fermentation tanks. Our research efforts will be directed in designing such control strategies.

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1. Introduction

Wine fermentation is much more complex than the simple transformation of must sugars into ethanol and CO₂. Many biological, chemical and physical processes occur simultaneously, yielding small amounts of hundreds of compounds that define most of the final organoleptic characteristics of the wine. Moreover, wine fermentation tanks are large; therefore temperature and concentration gradients can be significant. We believe that most troublesome fermentations are caused by extreme temperatures within the fermentation tank due to a poor temperature control system. Such problems can be reduced, for example, by using wine fermentation models, which are useful tools to assure wine quality and reproducibility among batches. Unfortunately, little progress has been made in this direction.

Most of the wine fermentation models developed until now, consider a homogenous liquid phase where basic biochemical phenomena associated to the yeast, such as main metabolism, growth limitation and inhibition, are described. Boulton (1979) presented one of the first models of this type, in which the fermentation rate depends on the temperature and the levels of glucose, fructose and ethanol. Later, Cramer et al. (2002) considered wine fermentation as nitrogen limited. However, the impact of nitrogen on biomass synthesis, metabolism activation and fermentation rate has only

been assessed recently (Varela et al., 2004; Coleman et al., 2007). Sainz et al. (2003) developed a dynamic flux balance analysis (DFBA) metabolic model for a nitrogen limited yeast alcoholic fermentation. In this model, fermentation kinetics is defined by the dynamic interaction between yeast main metabolism and extracellular conditions. The model was able to reproduce observed sugar consumption in both, nitrogen rich and nitrogen lean initial musts. DFBA models are appealing since parameters have a biological meaning, but their strong non-linearity can cause identifiability problems; therefore, sophisticated optimization algorithms are required to calibrate these models. Malherbe et al. (2004) developed a biochemical model for wine fermentation, where fermentation rate is defined by the population size and the number of sugar transporters, both of which were given by empirical non-linear functions of the initial assimilable nitrogen. Raghunathan et al. (2006) presented a specific and reliable algorithm to calibrate DFBA metabolic models and applied it to a variable structure yeast alcoholic fermentation model. The calibrated model reproduced biomass and sugar evolution along the fermentation. More recently, Pizarro et al. (2007) applied a DFBA model to describe anisothermal industrial red wine fermentations. This model was able to predict the end of the fermentations with an average error of 3.5 h.

Coleman et al. (2007) developed a predictive mathematical model for wine fermentation kinetics that includes the influence of temperature. The model was based on flask-scale wine fermentations at different temperatures and concentrations of nitrogen and sugar. According to the researchers, this is the first wine fermentation model that accurately predicted a transition from slug-

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gish to normal and to stuck fermentations as a function of temperature. Hitherto, no published wine fermentation model care about the inevitable temperature and concentration gradients that appear in large fermentation tanks. This is extremely relevant to ensure a favorable environment for the yeast during fermentation, since extreme temperatures and concentrations can be deleterious for its viability.

In this work, we developed a compartmental model to simulate the dynamics and temperature gradients of a 10 m³ wine fermentation tank. The model includes three compartments that exchange mass and energy. Each compartment considers biomass evolution, nutrient consumption (glucose, fructose and nitrogen), and production of ethanol and CO₂. The effects of pumping-over and heat exchange through cooling surfaces on temperature gradients are also included.

2. Materials and methods

2.1. Equipment description

The experimental data was obtained from a stainless steel fermentation tank (10 m³ of capacity, 3 m height, and 2.1 m internal diameter) equipped with two cooling jackets, one covering the external wall (8.9 m²) and the other placed at the bottom (3.4 m²). Additionally, a stainless steel cooling plate (1.9 m²) was installed on the central zone of the tank. Water flow rates to each cooling systems were handled by modulated solenoid valves. On-line instrumentation included three pressure sensors to estimate must height and density, four load cells to measure tank weight, three flow sensors to measure water cooling rates and six Pt100 temperature sensors. Five of these were located inside a vertical probe (40, 64, 210, 255 and 300 cm of height) and the other was placed next to the wall (35 cm deep). Data acquisition and control were handled with an OPTO22 system connected to a standard PC. Fig. 1 shows the piping and instrumentation diagram (P&ID) scheme of the fermentation tank.

2.2. Model description

The wine fermentation model presented below includes mass and energy balances in each compartment, mass and energy exchange between them, and rate expressions normally used in standard yeast fermentation models. The specific assumptions used in this model are:

- Nitrogen is the limiting substrate (Cramer et al., 2002).
- Sugar consumptions for ethanol and biomass production are independent (Cramer et al., 2002).
- Competitive inhibition is exerted between glucose and fructose, especially at high, total sugar concentration (Boulton et al., 1996).
- Ethanol inhibits sugar consumption.
- CO₂ accumulation in gas phase is negligible.
- Biomass viability depends on a combined effect of ethanol and temperature (Salmon, 2003; Sá-Correia and Van Uden, 1983).

2.3. Biomass

$$\frac{d(X_V \cdot H)}{dt} = (\mu - k_D) \cdot (X_V \cdot H) \quad (1)$$

where H is the must height in the fermenter and the growth rate, μ is given by,

$$\mu = \mu_{\max} \cdot \frac{N}{N + K_N(T)} \quad (2)$$

The cells death rate, k_D , is described by:

$$k_D = k_{D0} \cdot \exp \left[(C_{de} \cdot E) + E_{td} \frac{(T - 305.65)}{305.65 \cdot R \cdot T} \right] \quad T \geq T_D \quad (3)$$

$$k_D = 0 \quad T < T_D$$

In this expression, T_D represents the threshold temperature, from which thermal death is produced. This is associated to ethanol concentration (Salmon, 2003; Sá-Correia and Van Uden, 1983) according to the following polynomial:

$$T_D = -0.0001 \cdot E^3 + 0.0049 \cdot E^2 - 0.1279 \cdot E + 315.89 \quad (4)$$

Additionally, temperature affects the maximum growth rate according to (Boulton, 1979):

$$\mu_{\max} = \mu_0 \cdot \exp \left[\frac{E_{ac} \cdot (T - 300)}{300 \cdot R \cdot T} \right] \quad (5)$$

2.4. Nitrogen

$$\frac{d(N \cdot H)}{dt} = -\frac{\mu}{Y_{X/N}} \cdot (X_V \cdot H) \quad (6)$$

where $Y_{X/N}$ is the biomass/nitrogen yield coefficient.

2.5. Glucose and fructose

$$\frac{d(G \cdot H)}{dt} = -\left(\frac{\mu}{Y_{X/G}} + \frac{\beta_G}{Y_{E/G}} + m \cdot \frac{G}{G + F} \right) \cdot (X_V \cdot H) \quad (7)$$

$$\frac{d(F \cdot H)}{dt} = -\left(\frac{\mu}{Y_{X/F}} + \frac{\beta_F}{Y_{E/F}} + m \cdot \frac{F}{G + F} \right) \cdot (X_V \cdot H) \quad (8)$$

m is the specific cell maintenance rate and is weighted by the sugars content ratio (Boulton, 1979). β_G and β_F represent the ethanol production rates starting from glucose and fructose, respectively (Gee and Ramírez, 1994):

$$\beta_G = \beta_{G\max} \cdot \frac{G}{G + K_G(T)} \cdot \frac{K_{IE}(T)}{E + K_{IE}(T)} \quad (9)$$

$$\beta_F = \beta_{F\max} \cdot \frac{F}{F + K_F(T)} \cdot \frac{K_{IG}(T)}{G + K_{IG}(T)} \cdot \frac{K_{IE}(T)}{E + K_{IE}(T)} \quad (10)$$

These expressions are consistent with the sugar facilitated diffusion mechanism (Cramer et al., 2002; Malherbe et al., 2004).

Temperature affects the maintenance coefficient, according to the expression:

$$m = m_0 \cdot \exp \left[\frac{E_{am} \cdot (T - 293.3)}{293.3 \cdot R \cdot T} \right] \quad (11)$$

2.6. Ethanol

The classic Luedeking and Piret model without a growth associated factor is used here, since ethanol formation is low during the cell growth phase (Cramer et al., 2002; Pascal et al., 1995),

$$\frac{d(E \cdot H)}{dt} = (\beta_G + \beta_F) \cdot (X_V \cdot H) \quad (12)$$

The maximum ethanol production rate depends on temperature according to,

$$\beta_i = \beta_{i0} \cdot \exp \left[\frac{E_{afe} \cdot (T - 296.15)}{296.15 \cdot R \cdot T} \right] \quad i = G, F \quad (13)$$

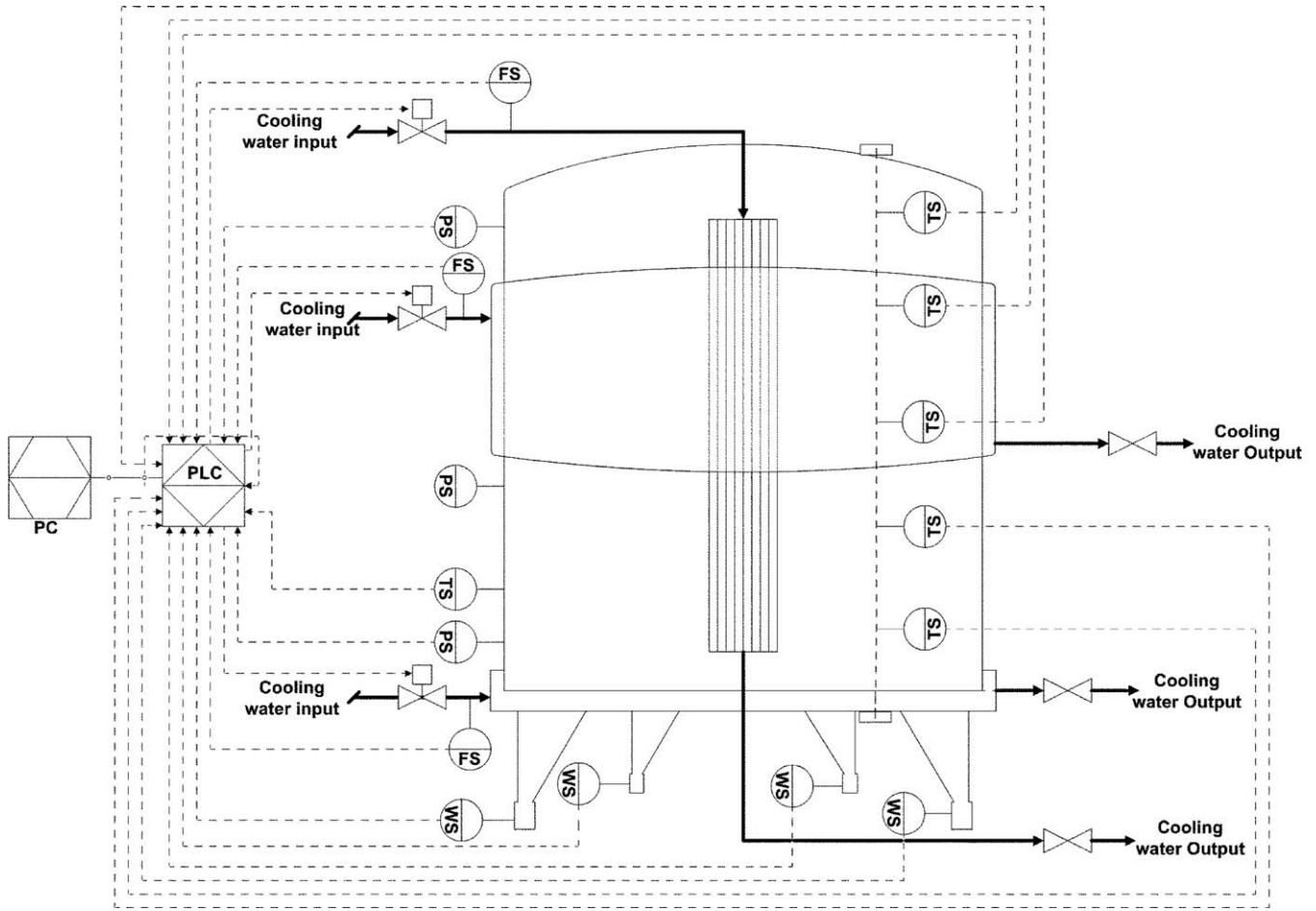


Fig. 1. P&ID of the fermentation tank.

2.7. CO₂

The CO₂ produced by the yeast is dissolved into the must until saturation and liberated through mass transfer to the gas phase (Gee and Ramírez, 1994). After saturation, all the CO₂ produced is released to the gas phase. Before saturation, the CO₂ accumulation in the must is modeled as:

$$\frac{d(C_{\text{CO}_2\text{L}} \cdot H)}{dt} = k_L \cdot a_{\text{TF}} \cdot H \cdot (C_{\text{sat}} - C_{\text{CO}_2\text{L}}) \quad (14)$$

The CO₂ production rate – contrary to ethanol – includes a growth associated factor since the highest production rates are registered during the cell growth phase (Malherbe et al., 2004; Sablayrolles and Barre, 1989). In this equation, $Y_{\text{X/CO}_2}$ and $Y_{\text{E/CO}_2}$ are the biomass/CO₂ and ethanol/CO₂ yield coefficients, respectively.

Since we assumed no accumulation in the gas phase, the flow rate of carbonic gas released to the atmosphere before saturation is given by:

$$F_{\text{CO}_2} = \left(\frac{\mu}{Y_{\text{X/CO}_2}} + \frac{\beta_G}{Y_{\text{E/CO}_2}} + \frac{\beta_F}{Y_{\text{E/CO}_2}} \right) X_v \cdot H \cdot A_F - k_L \cdot a_{\text{TF}} \cdot H \cdot (C_{\text{sat}} - C_{\text{CO}_2\text{L}}) \cdot A_F \quad (15)$$

where A_F is the cross-sectional area of the tank.

After saturation, no more CO₂ is dissolved into the must and all the gas produced by the yeast is released to the atmosphere, according to:

$$F_{\text{CO}_2} = \left(\frac{\mu}{Y_{\text{X/CO}_2}} + \frac{\beta_G}{Y_{\text{E/CO}_2}} + \frac{\beta_F}{Y_{\text{E/CO}_2}} \right) X_v \cdot H \cdot A_F \quad (16)$$

2.8. Total mass

This model considers that only CO₂ is released by the tank, according to:

$$\frac{d(\rho \cdot H)}{dt} = - \frac{F_{\text{CO}_2}}{A_F} \quad (17)$$

where ρ corresponds to the must density that has been correlated with temperature, and sugar and ethanol contents (Flood and Puagsa, 2000):

$$\rho = (176.2 \cdot W_a^2 - 59.24 \cdot W_e^2 + 50.34 \cdot W_a \cdot W_e + 361 \cdot W_a - 150.08 \cdot W_e + 998.69) \cdot f_c \quad (18)$$

$$f_c = -0.00075 \cdot T + 1.2242$$

2.9. Energy balance and temperature dependences

Assuming that the cooling system is on pseudo-steady state (very small volume), the outlet cooling water temperature is given by:

$$T_r = \frac{F_r \cdot c_{p_r} \cdot T_{\text{rin}} + U_R \cdot A_{\text{TCRF}} \cdot T + U_{\text{CRRA}} \cdot A_{\text{TCRA}} \cdot T_{\text{amb}}}{F_r \cdot c_{p_r} + U_R \cdot A_{\text{TCRF}} + U_{\text{CRRA}} \cdot A_{\text{TCRA}}} \quad (19)$$

The energy balance considers heat generation by fermentation, heat exchange through the cooling system, heat loss by CO₂ emission and the combined effect of environment convection and radiation,

$$\frac{d(\rho \cdot H \cdot (T - T^*))}{dt} = \frac{\Delta H}{c_{p_m}} \cdot \frac{d(S \cdot H)}{dt} - \frac{[U_R \cdot A_{TCRF} \cdot (T - T_r) + U_{CRFA} \cdot A_{TCFA} \cdot (T - T_{amb}) + F_{CO_2} \cdot c_{p_{CO_2}} \cdot (T - T^*)]}{A_F c_{p_m}} \quad (20)$$

where T^* is the reference temperature (20 °C), S is the total sugar concentration and ΔH is the heat released per sugar consumed.

All kinetic constants presented above depend on temperature according to the following Arrhenius-type equations (Boulton, 1979):

$$K_i(T) = K_{i0} \cdot \exp \left[\frac{E_{aki} \cdot (T - T^*)}{293.3 \cdot R \cdot T} \right] \quad i = N, G, F, IG, IE \quad (21)$$

2.10. Compartments and convective streams

It is empirically well known that the must temperature in the fermenting tank is not uniform. To approximate this observation, our model considers three compartments: upper-ring, upper-center and bottom. These compartments can exchange mass and energy, and each one is associated to one of the cooling system described above. Fig. 2 shows the compartments and the exchange streams.

During normal operation, exchange streams between compartments (F_{cv}) are caused by natural (density differences) and forced (mixing caused by CO₂ liberation) convection. In this model we assumed that the flow rates of these exchange streams were proportional to the temperature difference between the respective compartments:

$$\begin{aligned} F_{cv_{12}} &= F_{cv_{21}} = K_{cv_{12}} \cdot |\Delta T_{12}| \\ F_{cv_{13}} &= F_{cv_{31}} = K_{cv_{13}} \cdot |\Delta T_{13}|; \quad \text{if } T_3 > T_1 \\ F_{cv_{13}} &= F_{cv_{31}} = 0; \quad \text{if } T_3 \leq T_1 \\ F_{cv_{23}} &= F_{cv_{32}} = K_{cv_{23}} \cdot |\Delta T_{23}|; \quad \text{if } T_3 > T_2 \\ F_{cv_{23}} &= F_{cv_{32}} = 0; \quad \text{if } T_3 \leq T_2 \end{aligned} \quad (22)$$

During pumping-over, the exchange of mass and heat between compartments increases dramatically. This is modeled as three consecutive flow rate pulses (F_{rm}) in the exchange streams, each with a specified duration, magnitude and periodicity. Hence, the total ex-

change rate for a given component Z in a given compartment i is given by:

$$\begin{aligned} Ft_{i,Z} &= (F_{cv_{ij}} + F_{rm_{ij}}) \cdot \left[\frac{Z_j}{\rho_j} - \frac{Z_i}{\rho_i} \right] \\ &\quad + (F_{cv_{ik}} + F_{rm_{ik}}) \cdot \left[\frac{Z_k}{\rho_k} - \frac{Z_i}{\rho_i} \right] \\ i, j, k &= 1, 2, 3 \quad \text{with } j \neq i \text{ y } k \neq i, j \\ Z &= X_v, N, G, F, E, C_{CO_2L} \end{aligned} \quad (23)$$

Similarly, the total energy exchange for a given compartment i is given by:

$$\begin{aligned} Fe_i &= (F_{cv_{ij}} + F_{rm_{ij}}) \cdot c_{p_m} \cdot (T_j - T_i) \\ &\quad + (F_{cv_{ik}} + F_{rm_{ik}}) \cdot c_{p_m} \cdot (T_k - T_i) \\ i, j, k &= 1, 2, 3 \quad \text{with } j \neq i \text{ y } k \neq i, j \end{aligned} \quad (24)$$

In summary, in each compartment the model comprises the following mass and energy balances:

$$\frac{d(Z_i \cdot H_i)}{dt} = f_{bm_{i,Z}} + \frac{Ft_{i,Z}}{A_{Fi}} \quad (25)$$

$$\frac{d(\rho_i \cdot H_i)}{dt} = - \frac{F_{CO_{2i}}}{A_{Fi}} \quad (26)$$

$$\frac{d(\rho_i \cdot H_i \cdot (T_i - T^*))}{dt} = f_{be_i} + \frac{Fe_i}{c_p \cdot A_{Fi}} \quad (27)$$

$$i, j, k = 1, 2, 3 \quad \text{and} \quad Z = X_v, N, G, F, E, C_{CO_2L}$$

where $f_{bm_{i,Z}}$ represents the generation or consumption of Z in compartment i .

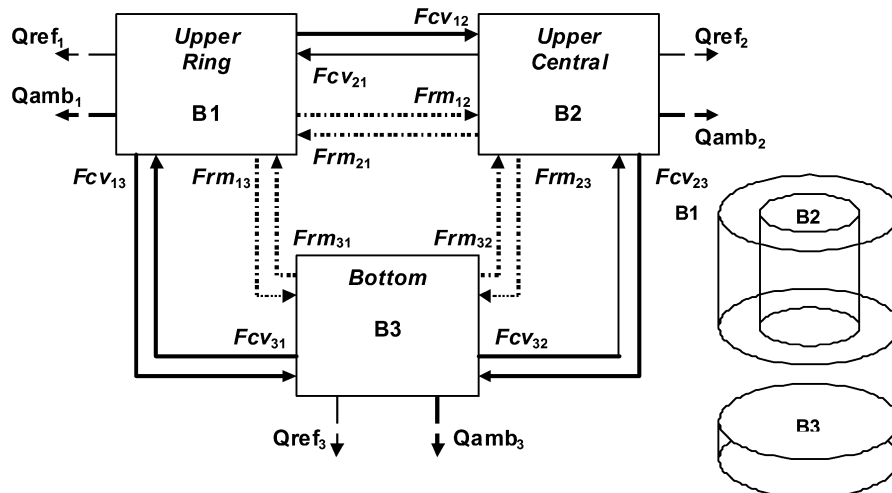


Fig. 2. Winemaking tank compartments and exchange streams.

Table 1
Nomenclature used in the model.

β_{i0}	Reference value of the specific production rate of ethanol ($i = G, F$), kg E/kg Bio/h	H_i	Height of compartment i in the tank ($i = 1, 2, 3$), m ³
β_i	Specific production rate of ethanol ($i = G, F$), kg E/kg Bio/h	$K_{cv_{ij}}$	Convection constant between compartment i and j ($i, j = 1, 2, 3$), kg/(h K)
$\beta_{\max}$	Maximum specific production rate of ethanol ($i = G, F$), kg E/kg Bio/hr	k_D	Specific death rate, 1/h
ΔH	Specific metabolic heat, kJ/kg S	k_{D0}	Reference value for the specific death rate, 1/h
μ	Specific growth rate, 1/h	K_F	Fructose saturation constant, kg F/m ³
μ	Reference value for the specific growth rate, 1/h	K_{F0}	Reference value for the fructose saturation constant, kg F/m ³
$\mu_{\max}$	Maximum specific growth rate, 1/h	K_G	Glucose saturation constant, kg G/m ³
ρ	Must density, kg/m ³	K_{G0}	Reference value for the glucose saturation constant, kg G/m ³
ρ_i	Must density in compartment i ($i = 1, 2, 3$), kg/m ³	K_{IE}	Ethanol inhibition constant, kg E/m ³
A_F	Fermentation tank base area, m ²	K_{IE0}	Reference value for ethanol inhibition constant, kg E/m ³
A_{Fi}	Base area of compartment i ($i = 1, 2, 3$), m ²	K_{IG}	Glucose inhibition constant, kg G/m ³
a_{TF}	Specific mass transfer area, m ² /m ³	K_{IG0}	Reference value for glucose inhibition constant, kg G/m ³
A_{TCFA}	External heat transfer area of the fermentation tank, m ²	k_{LA}	CO ₂ mass transfer coefficient, kg CO ₂ /(h m ² kg CO ₂ /m ³)
A_{TCRF}	Tank/cooling surface heat transfer area, m ²	K_N	Nitrogen compounds saturation constant, kg N/m ³
A_{TCRA}	Environment/cooling surface heat transfer area, m ²	K_{N0}	Reference value for nitrogen compounds saturation constant, kg N/m ³
C_{CO_2L}	Dissolved CO ₂ , kg CO ₂ /m ³	m	Maintenance coefficient, kg S/kg Bio/h
C_{de}	Ethanol death constant, m ³ /kg E	m_0	Reference value for maintenance coefficient, kg S/kg Bio/h
cp_r	Specific heat of cooling fluid, kJ/kg/K	N	Nitrogen compounds concentration, kg N/m ³
cp_{CO_2}	CO ₂ specific heat, kJ/kg/K	R	Ideal gas constant, kJ/kmol/K
cp_m	Must specific heat, kJ/kg/K	S	Total sugar concentration, kg S/m ³
C_{sat}	Saturation of dissolved CO ₂ , kg CO ₂ /m ³	t	Time, h
E	Ethanol concentration, kg E/m ³	T	Must temperature, K
E_{ac}	Activation energy of growth kinetics, kJ/kmol	T_i	Must temperature in compartment i ($i = 1, 2, 3$), K
E_{afe}	Activation energy of ethanol production rate, kJ/kmol	T^*	Reference temperature, K
E_{akf}	Activation energy of saturation fructose constant, kJ/kmol	T_D	Temperature of thermal death, K
E_{akg}	Activation energy of saturation glucose constant, kJ/kmol	T_r	Cooling fluid outlet temperature, K
E_{akie}	Activation energy of inhibition ethanol constant, kJ/kmol	T_{rin}	Cooling fluid inlet temperature, K
E_{akig}	Activation energy of inhibition glucose constant, kJ/kmol	T_{amb}	External temperature, K
E_{akn}	Activation energy of saturation nitrogen compounds constant, kJ/kmol	U_{CRRRA}	Global heat transfer coefficient (convection and radiation) between cooling system and the environment, kJ/(m ² h K)
E_{am}	Activation energy of maintenance, kJ/kmol	U_{CRFA}	Global heat transfer coefficient (convection and radiation) between fermentation tank and the environment, kJ/(m ² h K)
E_{td}	Activation energy of thermal death, kJ/kmol	U_R	Global heat transfer coefficient between cooling system and fermentation tank, kJ/(m ² h K)
F	Fructose concentration, kg F/m ³	W_a	Sugars mass fraction (S/ρ), kg S/kg must
$f_{bm_{i,Z}}$	Generation or consumption of Z compound in compartment i ($Z = X, N, G, F, E, C_{CO_2L}$; $i = 1, 2, 3$)	W_e	Ethanol mass fraction ($E/(\rho - S)$), kg E/kg must _{sugar free}
f_{be_i}	Heat exchange into compartment i ($i = 1, 2, 3$)	X_v	Biomass concentration, kg Bio/m ³
f_c	Correction factor of density by temperature	X_{vf}	Biomass concentration at the end of the fermentation, kg Bio/m ³
F_{CO_2}	Mass flow rate of liberated CO ₂ , kg CO ₂ /h	$Y_{X/N}$	Biomass/nitrogen yield coefficient, kg Bio/kg N
$F_{cv_{ij}}$	Exchanged mass by natural convection between compartments i and j ($i, j = 1, 2, 3$)	$Y_{X/G}$	Biomass/glucose yield coefficient, kg Bio/kg G
Fe_i	Exchanged energy into compartment i ($i = 1, 2, 3$)	$Y_{X/F}$	Biomass/fructose yield coefficient, kg Bio/kg F
F_r	Mass coolant flow, kg/h	$Y_{E/G}$	Ethanol/glucose yield coefficient, kg E/kg G
$F_{rm_{ij}}$	Exchanged mass by pumping-over between compartments i and j ($i, j = 1, 2, 3$)	$Y_{E/F}$	Ethanol/fructose yield coefficient, kg E/kg F
$F_{t_{i,Z}}$	Exchanged mass of Z compound into compartment i ($i = 1, 2, 3$)	Y_{X/CO_2}	Biomass/CO ₂ yield coefficient, kg Bio/kg CO ₂
G	Glucose concentration, kg G/m ³	Y_{E/CO_2}	Ethanol/CO ₂ yield coefficient, kg E/kg CO ₂
H	Height of the fermentation tank, m ³		

2.11. Model parameters

Table 1 shows the nomenclature used in this article and Table 2 exhibits the model parameters and constants (most of them were taken from the literature). Table 3 shows a list of the parameters of each compartment of the tank.

Simulations were performed on a standard PC using MATLAB®/SIMULINK environment and the *ode45* (Dormand-Prince) integration routine.

3. Results and discussion

3.1. Comparison of experimental and simulated results

It was hard to compare simulated and measured temperatures in the experimental tank. While simulated temperatures represented the average of the respective zone in the tank, experimental temperatures gave measurements at specific points within the fermentor. Hence, measured temperatures that were inside the corresponding tank zones defined in the model, were compared with the respective simulated temperatures. The two upper sensors in the vertical probe were located outside the must, and were not considered in this analysis. Thus, the measured temperatures considered for comparison purposes were three in the vertical probe: TS1 (40 cm height), TS2 (164 cm height) and TS3 (210 cm height); and one attached to the wall (35 cm height), named TP1. A unique

feature of our model is the ability to compute three different temperatures, according to the modeled compartment: T_1 (upper-ring temperature), T_2 (upper central temperature) and T_3 (bottom temperature). The following equivalence between measured and simulated temperatures was considered (see Table 4).

Given the location of the probes in the experimental tank, the simulated temperature of the upper-ring compartment was not able to be verified experimentally. Therefore, we compared the measured temperatures with the simulated ones in the central and bottom compartments only. Fig. 3a and b shows that the model was able to reproduce the dynamics of measured temperatures at different places of the tank during 200 h of recorded fermentation; with an absolute mean error of 1.2 °C and 1.4 °C, respectively. Water flow rates for each cooling system were defined by recorded ON/OFF operating intervals of the cooling valve in the experimental fermentor; this data was used as input variables in the simulations. The peaks observed in the figures were caused by the effect of pumping-over, which is extensively used in the wine making process as a way to avoid excessive high temperatures within the fermentor. Another unique feature of our model is that it can simulate the pumping-over effect, by allowing intensive mass and energy exchange between the model compartments.

The maximum temperature difference registered by the RTD's was 5.8 °C (TS1 and TS2), and was reached at approximately 61 h of fermentation. For the same compartments (central and bottom), the model registered a maximum temperature difference of 5.2 °C

Table 2
Values of parameters and constants used in the model.

Parameter or constant	Value	Reference
β_{FO}	0.225 kg E/kg Bio/h (23 °C)	Estimated from bs value and considering a preferred consumption of glucose over fructose of 3:1 (Cramer et al. 2002; Salmon 2003)
β_{GO}	0.225 kg E/kg Bio/h (23 °C)	
C_{de}	0.0415 m ³ /kg E	Adjusted to data obtained by Sá-Correia and Van Uden (1983), choosing values that provide a reasonable prediction of the final biomass at the end of fermentation ($X_{vf} = 3$ g Bio/L) (Salmon 2003)
E_{td}	130,000 kJ/kmol	
k_{DO}	0.00044 1/h	
CP_m	4.15 kJ/kg/K	Average must and wine (Boulton et al., 1996)
CP_r	4.19 kJ/kg/K	Boulton et al. (1996)
ΔH	−651.28 kJ/kg	Boulton (1979)
E_{ac}	59,453 kJ/kmol	Boulton (1979)
E_{afe}	11,000 kJ/kmol	Estimated by trial and error
E_{akf}	46,055 kJ/kmol	Boulton (1979)
E_{akg}	46,055 kJ/kmol	Boulton (1979)
E_{akie}	46,055 kJ/kmol	Boulton (1979)
E_{akig}	46,055 kJ/kmol	Boulton (1979)
E_{akn}	46,055 kJ/kmol	Boulton (1979)
E_{am}	37,681 kJ/kmol	Boulton (1979)
K_{FO}	7.5 kg F/m ³ (24 °C)	Estimated from bs value and considering a preferred consumption of glucose over fructose of 3:1, without inhibitory effects and considering $K_S = 15$ kg S/m ³ (Cramer et al. 2002; Salmon 2003; Malherbe et al. 2004)
K_{GO}	7.5 kg G/m ³ (24 °C)	
K_{JO}	40 kg E/m ³ (20 °C)	Boulton (1979)
K_{JO}	5/4 kg G/m ³	Estimated considering a preferred consumption of glucose over fructose of 3:1 (Salmon 2003)
$k_L \cdot a_{TF}$	0.07 1/h	Gee and Ramirez (1994)
K_{NO}	0.01 kg N/m ³ (23 °C)	Cramer et al. (2002)
μ	0.18 1/h (26.85 °C)	Boulton (1979)
m_0	0.01 kg S/kg Bio/h (20 °C)	Boulton (1979)
R_*	8.314 kJ/kmol/K	Boulton et al. (1996)
T_*	293.15 K	
T_{amb}	295.15 K	Experimental data (average value)
T_{rin}	285.15 K	Experimental data (average value)
X_i	0.18 kg/m ³	Experimental data
Y_{E/CO_2}	1.05 kg E/kg CO ₂	Estimated stoichiometrically from a reaction system formed by biomass (CH _{1.66} N _{0.16} O _{0.5}), glucose, fructose, ammonia, ethanol, CO ₂ and H ₂ O, (Pascal et al., 1995) without ethanol production associated to growth.
Y_{X/CO_2}	12.72 kg Bio/kg CO ₂	
$Y_{X/F}$	1.60 kg Bio/kg F	
$Y_{X/G}$	1.60 kg Bio/kg G	
$Y_{E/F}$	0.49 kg E/kg F	Estimated stoichiometrically from a reaction system formed by biomass (CH _{1.66} N _{0.16} O _{0.5}), glucose, fructose, ammonia, ethanol, CO ₂ and H ₂ O, (Pascal et al., 1995) without ethanol production associated to growth. Weighted by 0.95
$Y_{E/G}$	0.49 kg E/kg S	Estimated from a non-linear correlation, considering an average biomass of 3·10 ^{−11} g/cell, an initial population of 10 ^{−12} cell/m ³ and an initial nitrogen content of assimilable nitrogen of 0.28 kg/m ³ (Malherbe et al., 2004)
$Y_{X/N}$	19.69 kg Bio/kg N	

Table 3
Values of characteristic parameters of each compartment.

Parameter	Compartment 1 (ring)	Compartment 2 (superior-center)	Compartment 3 (bottom)	Reference
Refrigeration System	Jacket	Immersion plate	Jacket	
A_{TCRA}	9.23 m ²	–	3.46 m ²	Physical dimension of the tank
A_{TCRF}	9.23 m ²	1.88 m ²	3.46 m ²	Physical dimension of the tank
H_{max}	1.4 m	–	1.9 m	Physical dimension of the tank
r_{int}	1.05 m	–	1.05 m	Physical dimension of the tank
Ring height	1.44 m	–	–	Physical dimension of the tank
U_R	180 kJ/(m ² h K)	50 kJ/(m ² h K)	300 kJ/(m ² h K)	Trial and error by simulation
U_{CRRA}	33 kJ/(m ² h K)	–	43 kJ/(m ² h K)	Trial and error by simulation

Table 4
Model compartments and equivalent probes in the experimental tank.

Compartment	Experimental temperature	Simulated temperature
Upper-ring	–	T1
Central	TS2, TS3	T2
Bottom	TS1, TP1	T3

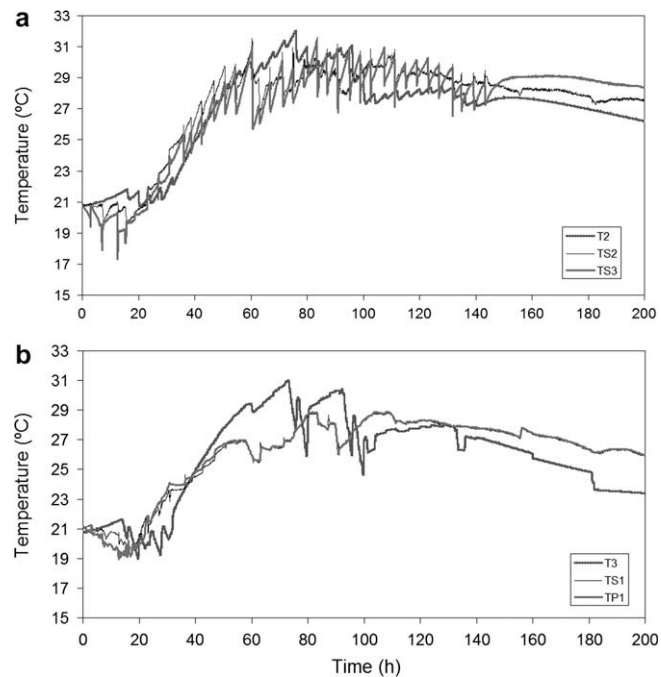


Fig. 3. Temperatures simulated by the model and temperatures measured in the experimental tank in: (a) central compartment, (b) bottom compartment.

(between T2 and T3). In this simulation, temperature differences greater than 5 °C were observed after 80 h of simulated fermentation. We would like to stress here that the model computes average temperatures in the compartments, and not at specific points as recorded by RTDs TS1, TS2, TS3 and TP1.

Fig. 4 shows the response of the simulated temperatures in the central compartment of the model, with and without a cooling system. This compartment was chosen over the others, because the maximum recorded temperatures are reached here. Additionally, we simulated the same cooling operating conditions found in our experimental tank (Fig. 3), including the same pumping-over intervals. When the cooling system was disconnected, the only mechanism that the system had available to cool the must was heat convection with the ambient, and heat exchange with the other compartments (pumping-over). In this case, the maximum temperature almost reached 36.5 °C at approximately 63 h of fermenta-

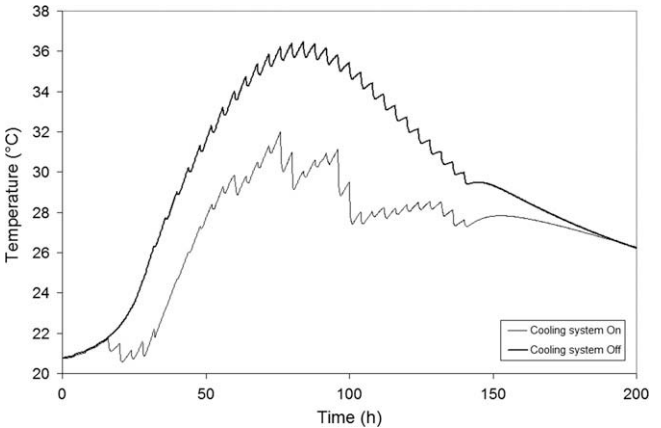


Fig. 4. Simulated temperatures of central compartment with and without cooling system.

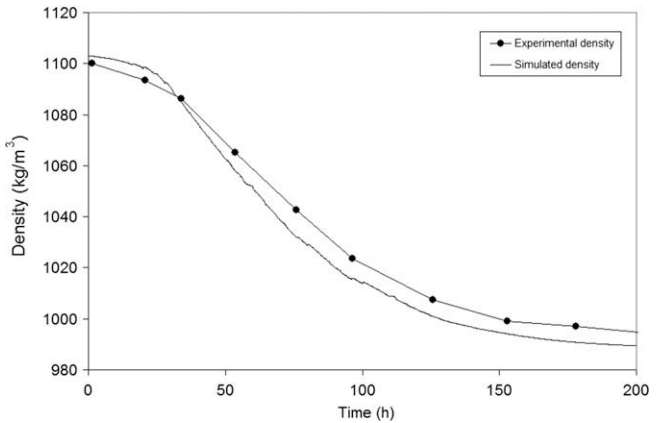


Fig. 5. Evolution of experimental and simulated densities.

tation; such temperatures can be easily found in uncontrolled industrial fermentations.

Fig. 5 compares simulated (average of the three zones) and measured densities. Here, the dynamics of the simulated density is very similar to the measured one; with an offset no greater than 5 kg/m³. The distance between the curves can be easily reduced by fitting more model parameters, but this is not necessary for designing cooling and control systems, case where the dynamics is more important.

Typical curves of active biomass and ethanol evolution are shown in Fig. 6a. Active biomass is hard to measure directly, because of the difficulty to separate active from inactive cell mass. However, if sluggish or stack fermentations are to be predicted, the use of active biomass – as opposed to total biomass – is crucial.

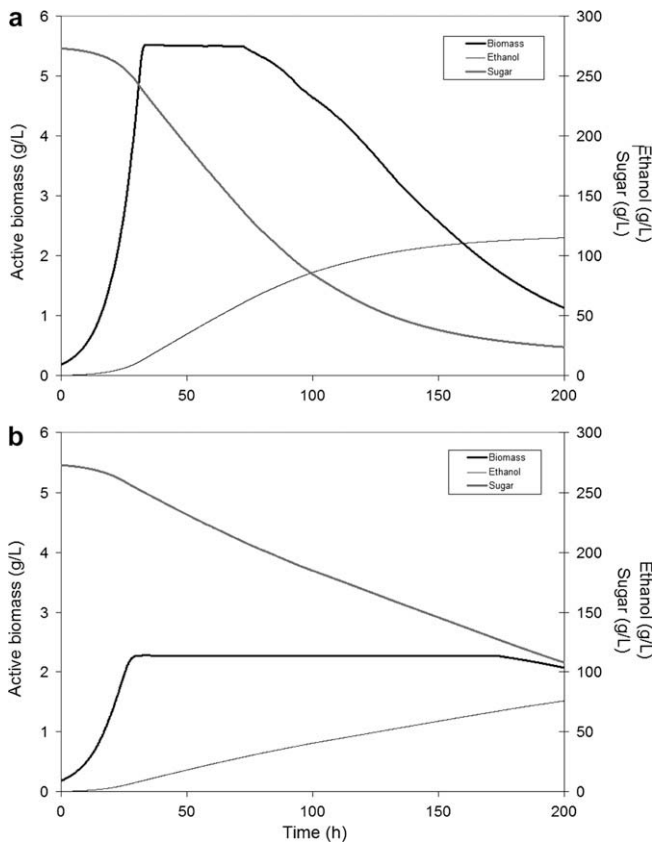


Fig. 6. Comparison of: (a) a normal fermentation and (b) a sluggish fermentation caused by a 70% reduction in the initial amount of nitrogen.

Table 5

Average and highest temperature differences due to variations in $\pm 10\%$ of selected parameters.

Parameter	$ \Delta T_{ave} $	$ \Delta T_{Max} $
U_{R1}	0.11	0.37
U_{R3}	0.19	0.65
T_{amb}	1.40	2.20
T_{rin}	0.12	0.36

Fig. 6a shows the evolution of active biomass and ethanol concentration. As expected, the model computed a final ethanol concentration of 115 g/L, very similar to the values found in industrial wine fermentations. Additionally, maximum biomass concentrations were 5.5 g/L, and after 200 h of simulated fermentation, sugar was almost consumed reaching 23 g/L. In a recent study, Coleman et al. (2007) successfully predicted sluggish and stuck fermentations by manipulating culture temperature, nitrogen levels and initial sugar concentrations. As stated by Coleman et al. (2007), sugar utilization is of the utmost importance, because of its value to predict a stuck or sluggish fermentation. In order to test our model, we reduced in 70% the initial concentration of nitrogen and kept all other values constant, including cooling (Fig. 6b). In Fig. 6b, maximum biomass concentrations were 2.3 g/L, and after 200 h of fermentation, our model predicted high levels of sugar (108 g/L); a clear indication of the presence of a sluggish fermentation.

3.2. Sensitivity analysis of the simulated temperatures

The ultimate goal of our model is to design strategies for must temperature control during fermentation. In order to assess the

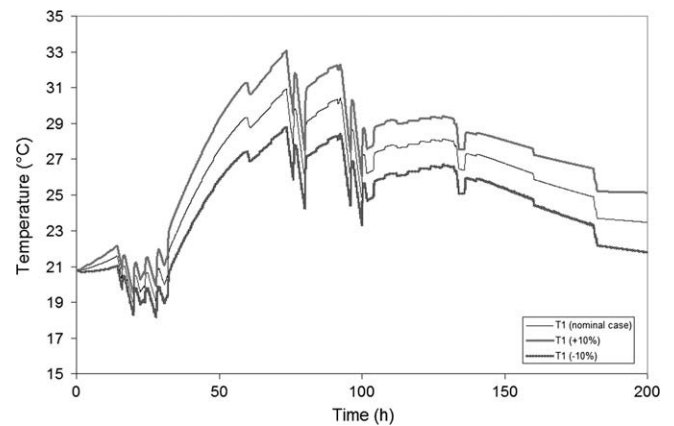


Fig. 7. Temperature dynamics of the bottom compartment considering variations of $\pm 10\%$ in the ambient temperature.

uncertainty associated with several of the model's parameters, a sensitivity analysis was performed. The value of each parameter that was not obtained from literature and did not correspond to a physical dimension was modified in $\pm 10\%$. Table 5 shows the average and highest temperature difference along with the most significant parameters.

The most important temperature differences (2.2 °C) were observed when the ambient temperature was perturbed in $\pm 10\%$ (Table 5). For the other cases, temperature differences did not exceed, in average, more than 0.2 °C with respect to the nominal case. This is a positive feature of the model, because ambient temperature can be measured with high precision, and the impact of other parameter's uncertainty is not significant. Hence, for illustrative purposes, Fig. 7 shows $\pm 10\%$ variations in the ambient temperature and its effect in the bottom compartment of the model.

4. Conclusions

The model presented in this article represents well the complexity of the temperature dynamics observed in a wine fermentation tank, aspect not considered by previous models. For example, simulated maximum temperatures differences were similar to those recorded in an experimental tank. When fermentations were simulated without a cooling system, excessive temperatures were obtained. Additionally, when low initial concentrations of nitrogen were used, sluggish fermentations were predicted, as would be expected in a real wine fermentation with deficient nitrogen concentrations.

The model can be easily adapted to design and assess control strategies to avoid extreme temperatures in large wine fermentation tanks, where temperature gradients are more significant. Our future efforts will be directed in designing such control strategies.

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